Foreword

In this document I explain in a few easy steps on how to setup the windows virtual machine on p3ng0s and how to use it accordingly for building C# programs and cloning a windows environment from an exploited machine. You will also be able to boot from a plugged in USB drive.

Setup

First start in `loot/` to create a folder named qemu and place in it the following `config.json` file:

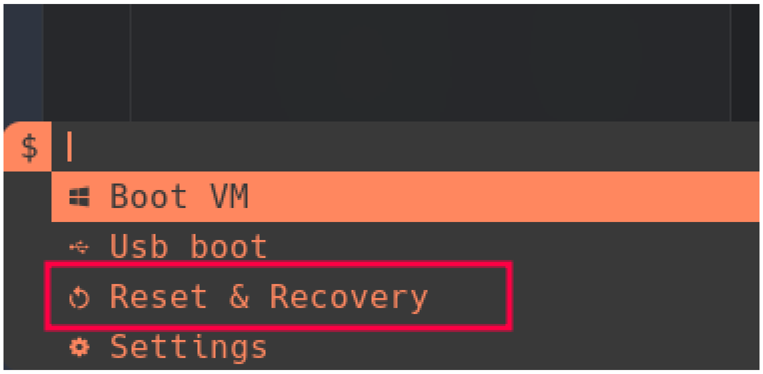
```
{
    "ram":"8G",
    "cores":5,
    "vm_file":"/home/p4p1-live/loot/qemu/Windows.img",
    "vm_size":"100G",
    "virtio":"/home/p4p1-live/loot/qemu/virtio-win.iso",
    "iso":"/home/p4p1-live/loot/qemu/windows.iso",
    "scripts":"/home/p4p1-live/loot/qemu/scripts/"
    }
```

Requirements

- Windows Virtio-Drivers
- windows 10 ISO
- qemu packages

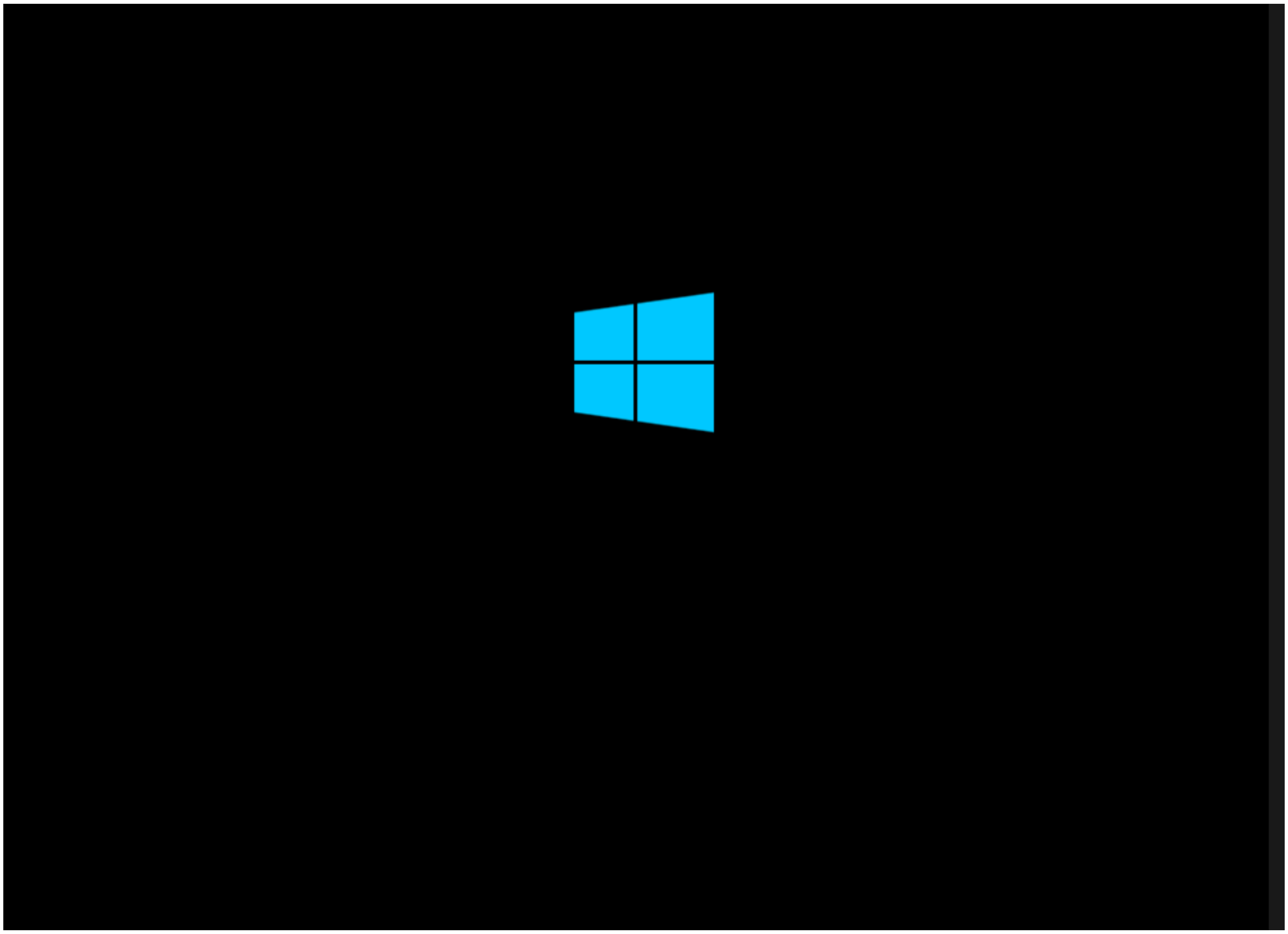
Install of windows

The installation of windows is quite straight forward and follows the same principle as any other installs. As long as you have the correct files in the correct folder everything should work fine! Inside of your $HOME folder you should save the Windows.iso file as the following: `~/.qemu/windows.iso`. After that you can run the dmenu_win script by pressing the MOD+v (By default MOD -> Alt) shortcut on your keyboard:



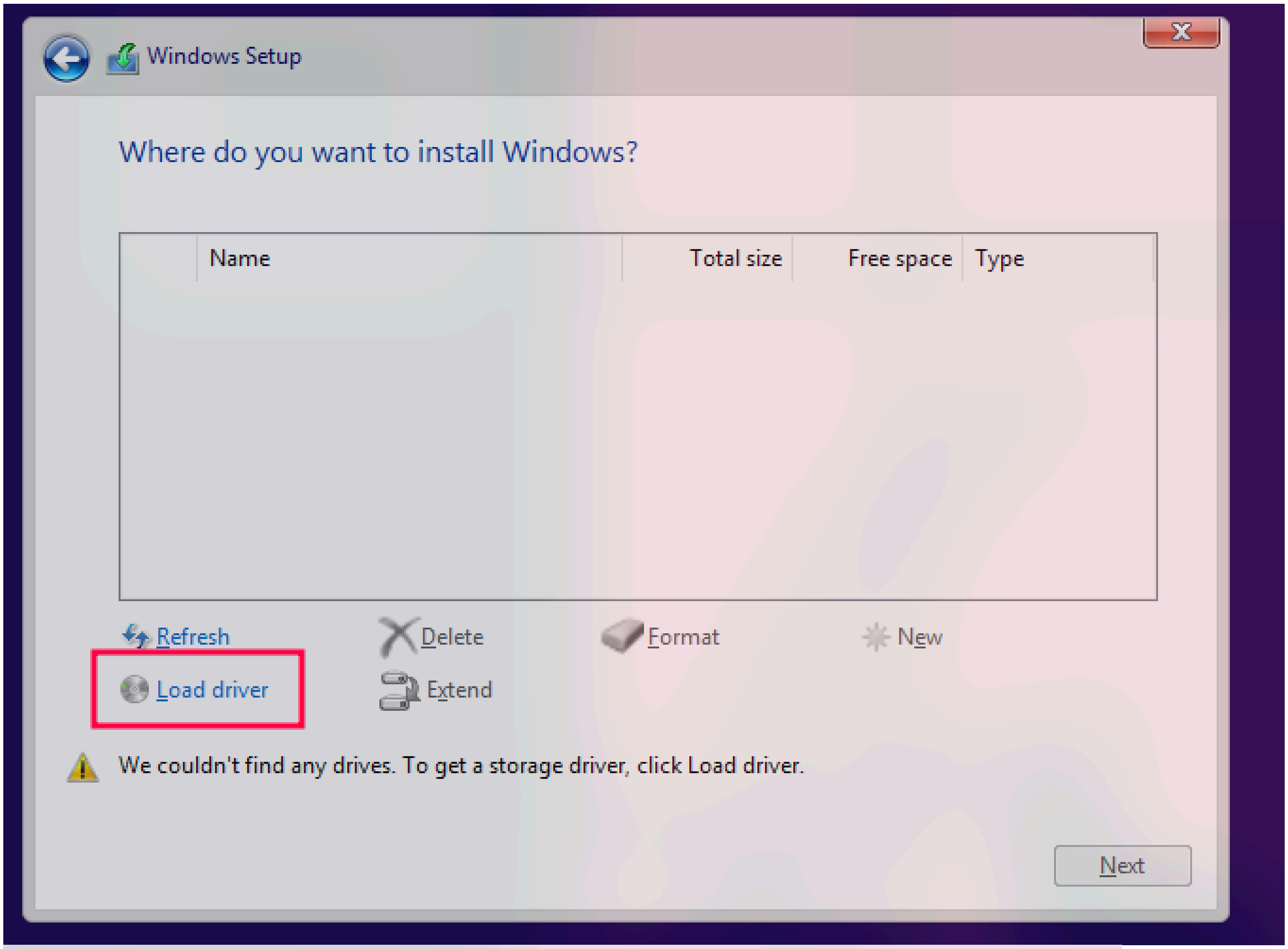
From there you can just proceed by selecting the Reset & Recovery option which will launch the creation with the .iso loaded. Don't forget to press Escape to view the boot menu:

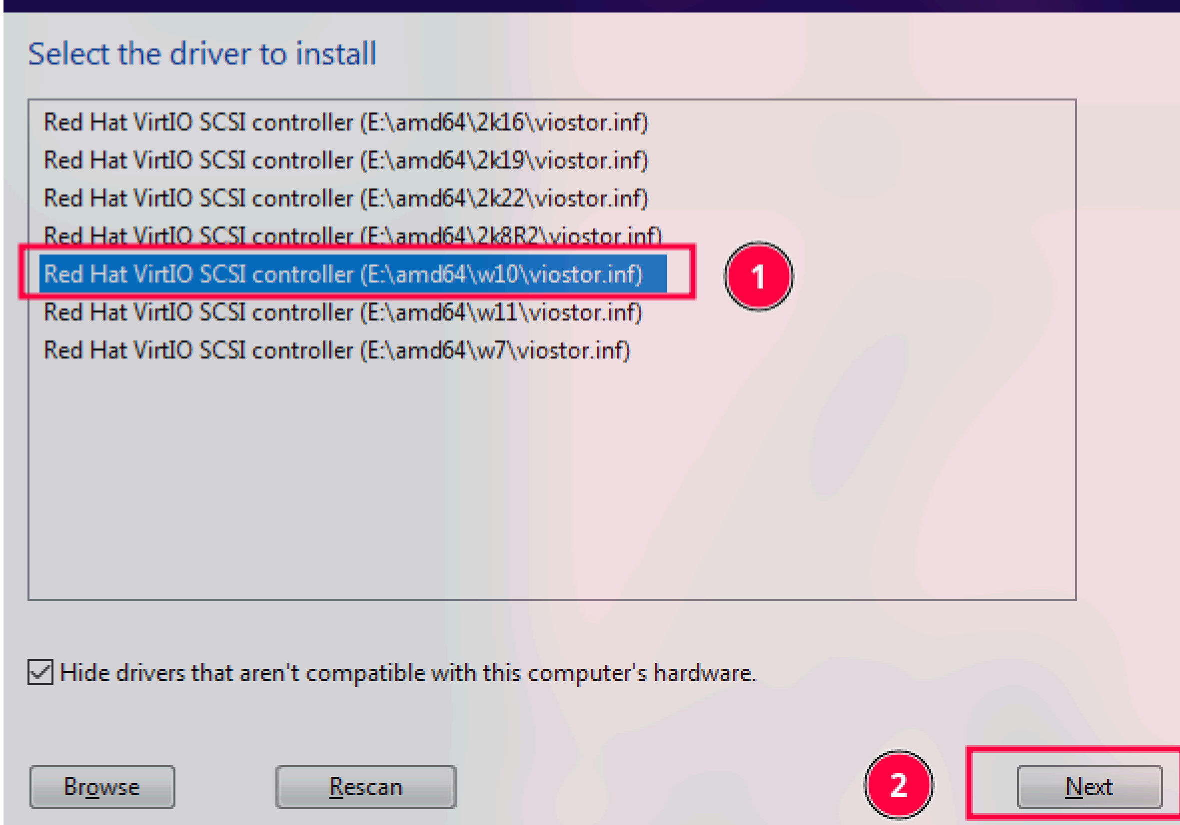
You should now be booted inside of the normal windows ISO install! It does take a bit of time to boot on first setup so if you get a blinking cursor at the top of the screen just wait for about 10 minutes to be sure it isn't just slow :)



You might need to do the following for windows 11:

Inside of the live windows.iso make sure you install the Correct drivers through this dialog since you won't see any hard drives:

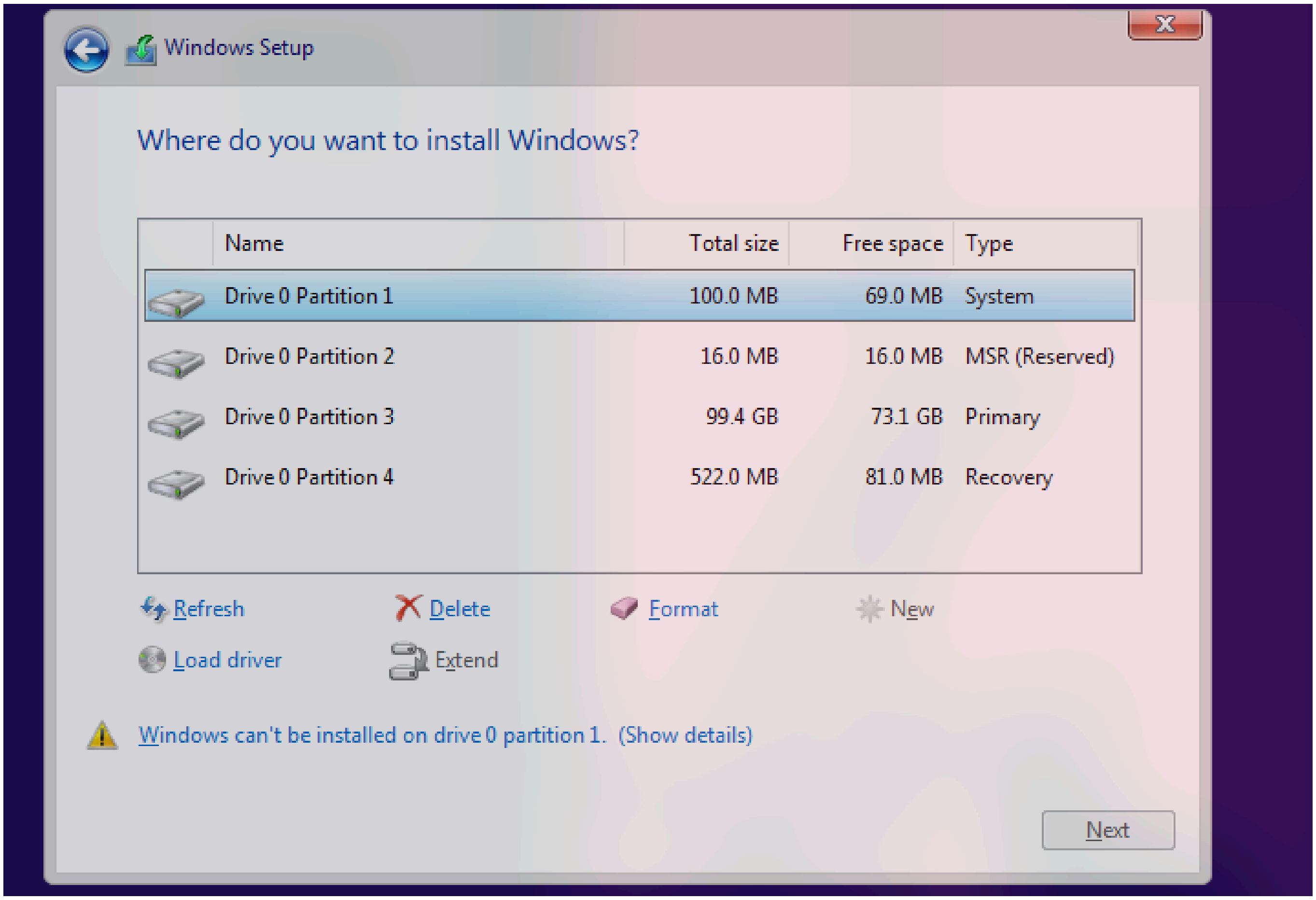




You can also install the drivers with the following commands:

```
wmic logicaldisk get caption,volumename
pnputil /add-driver D:\viostor\w11\amd64\*.inf /install
```

Once the drivers installed you should see the drives to install windows on:



Note

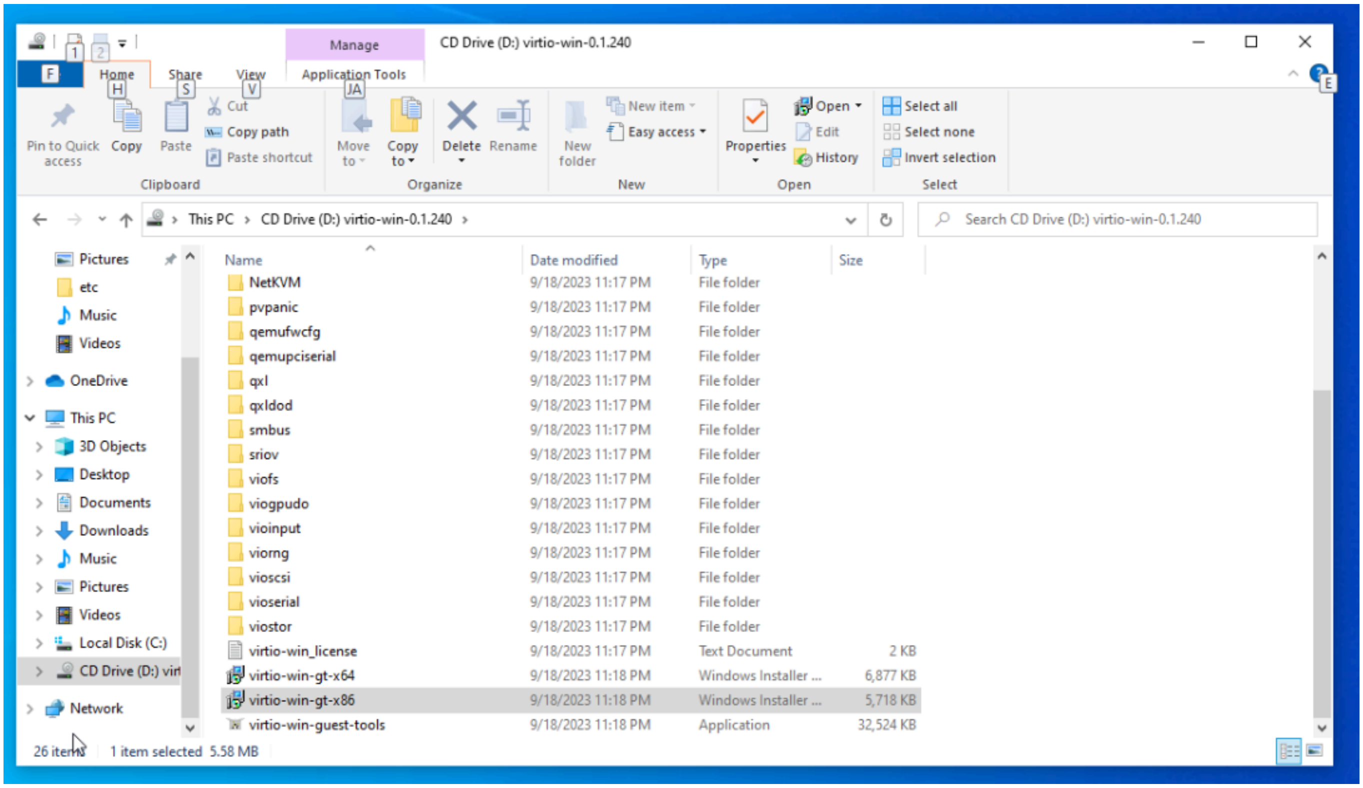
No drivers will be installed so make sure you install all of the drivers after reboot!

Once you are going through the windows install if it prompts you for Credentials type in the following:

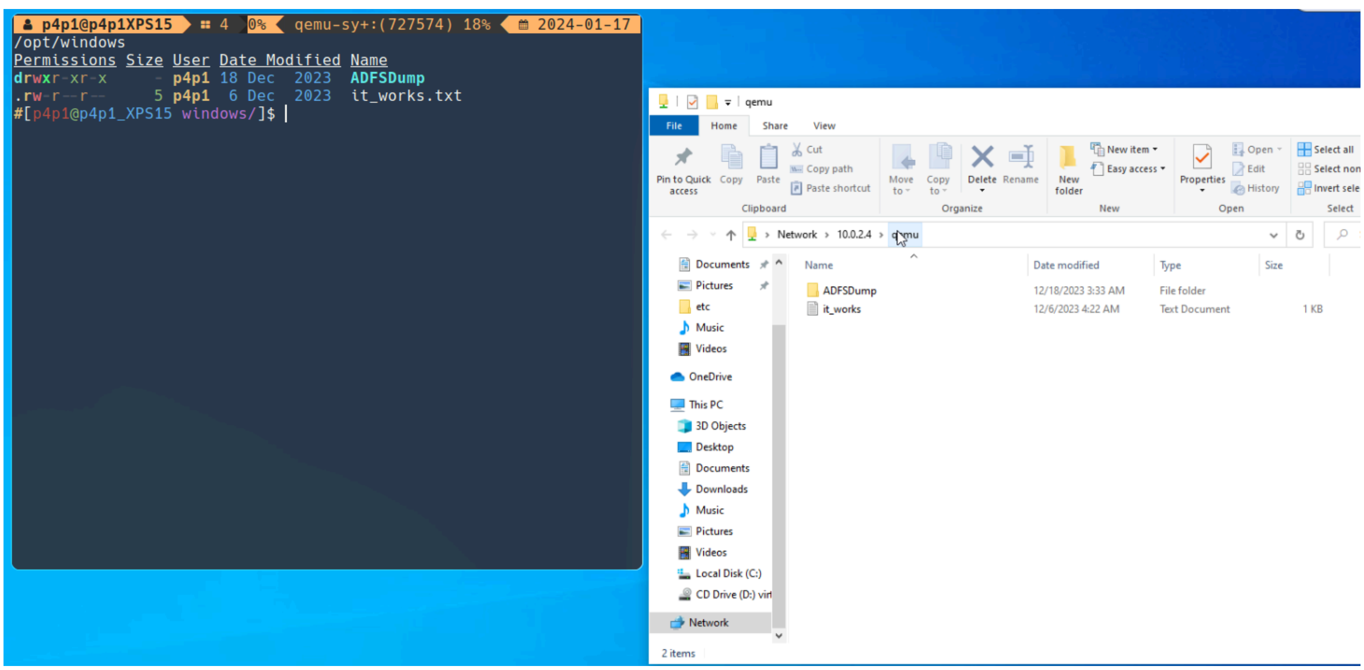
```
no@thankyou.com:fkjdlas
```

Drivers and shared folders

Now you must install drivers for maximum support with your machine and just so that everything runs smoothly. To do so login to your windows VM and select the following:



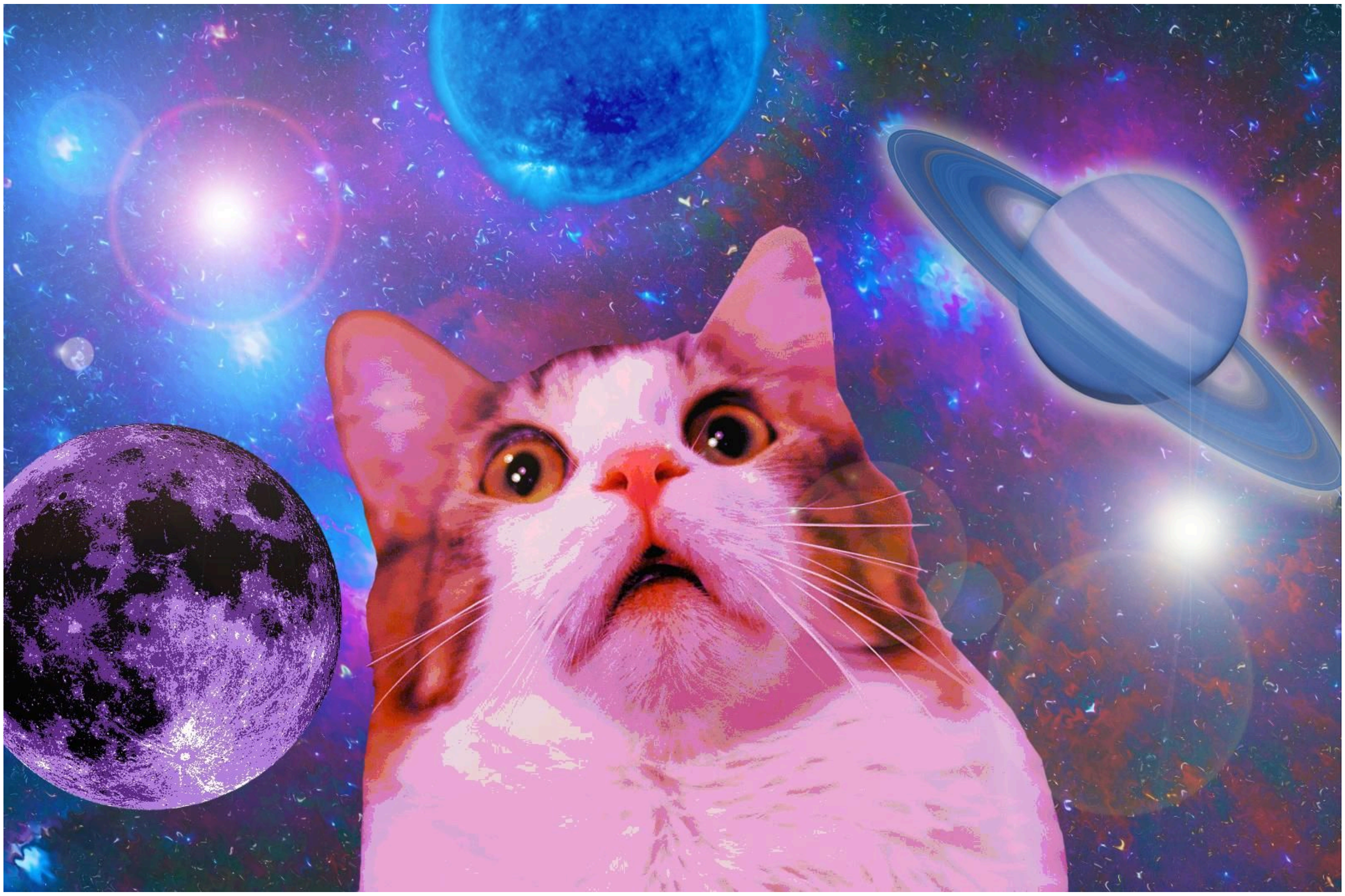
The D drive should have the virtio drivers and you can run the `virtio-win-gt-x64.exe` file! The shared folder is made as a SMB file share located on the following link: `\\10.0.2.4\qemu`

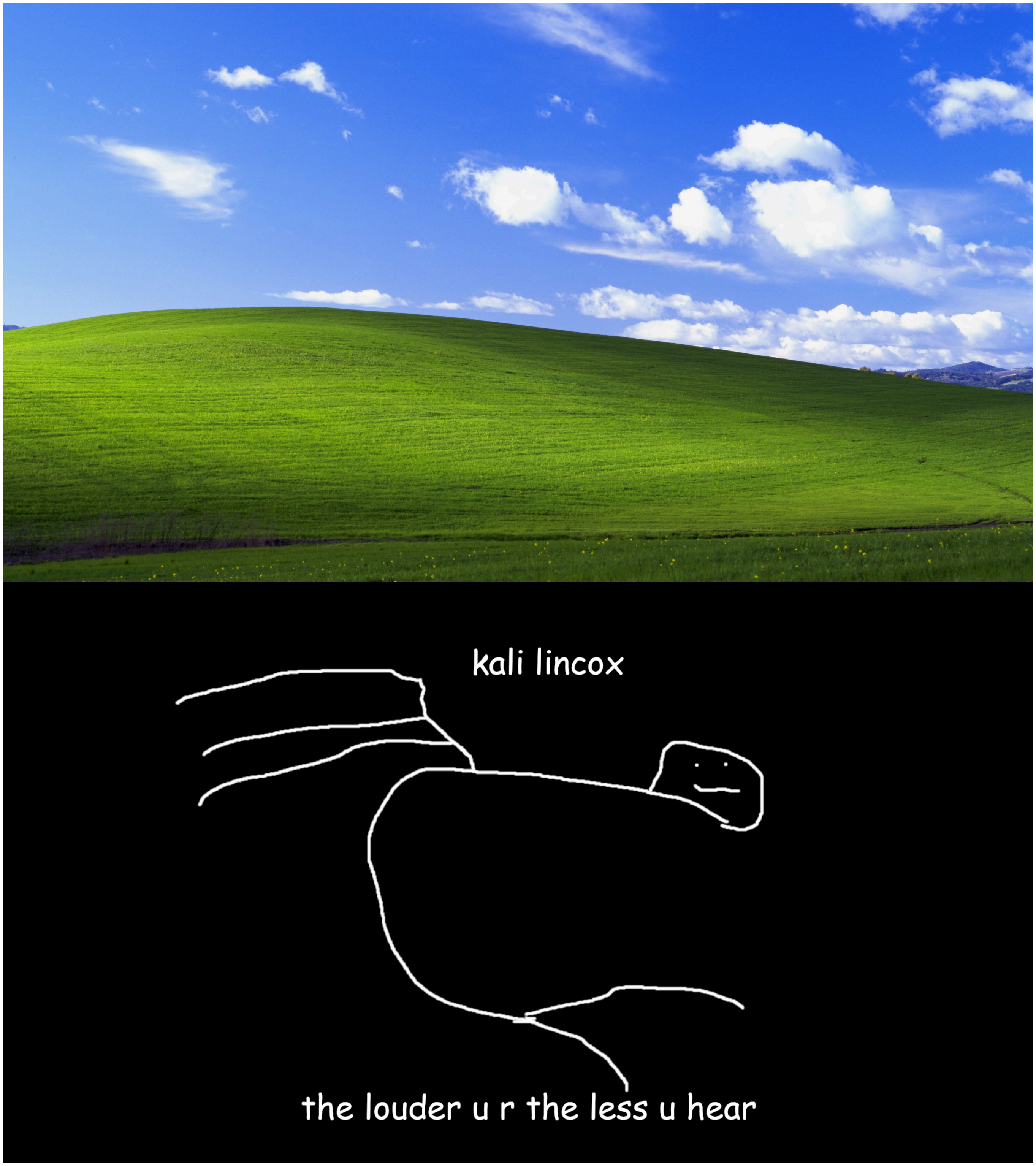


That is directly mapped to the /opt/windows/ folder inside of your p3ng0s installation!

Windows wallpaper

Here is my windows wallpapers

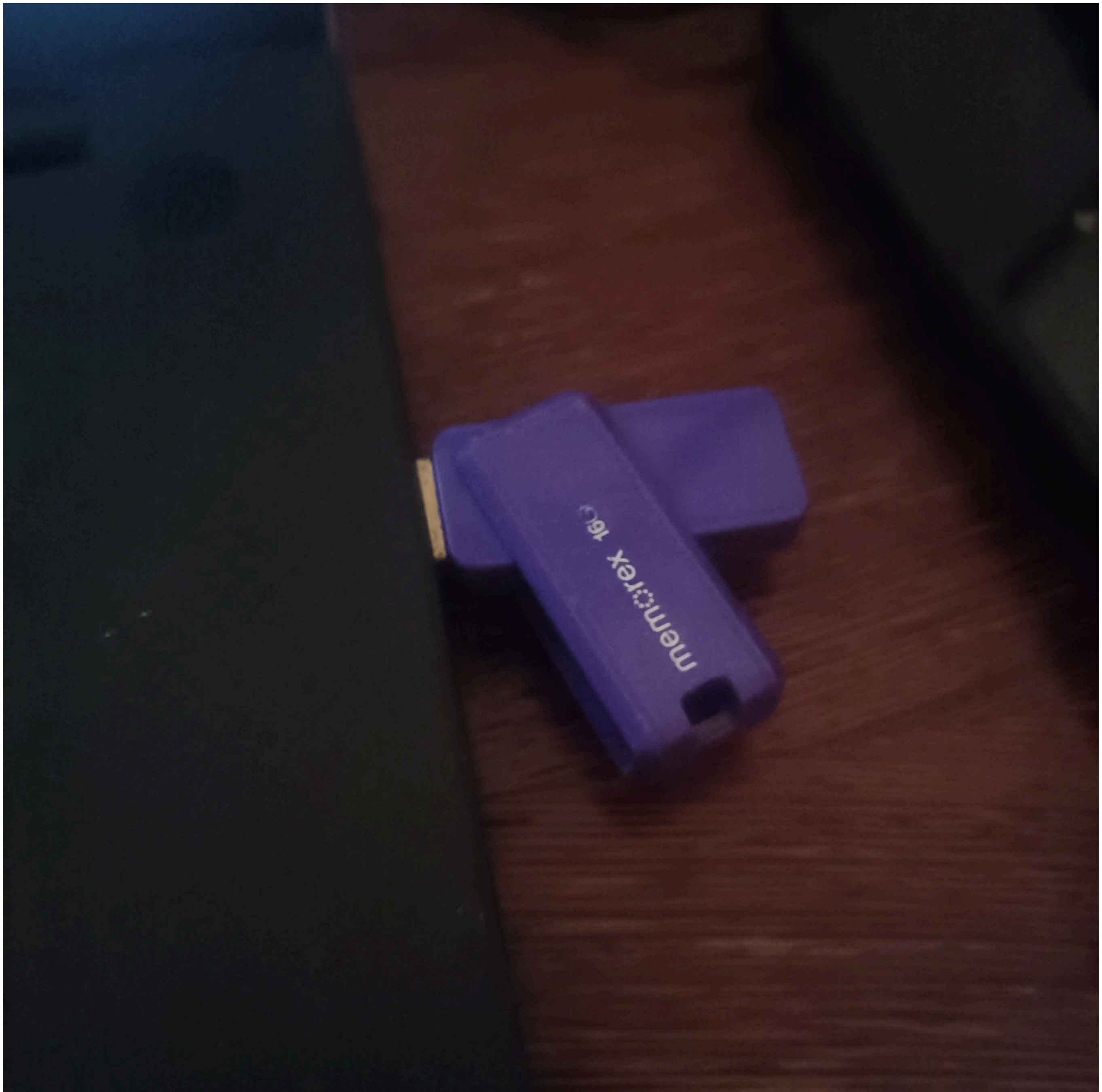


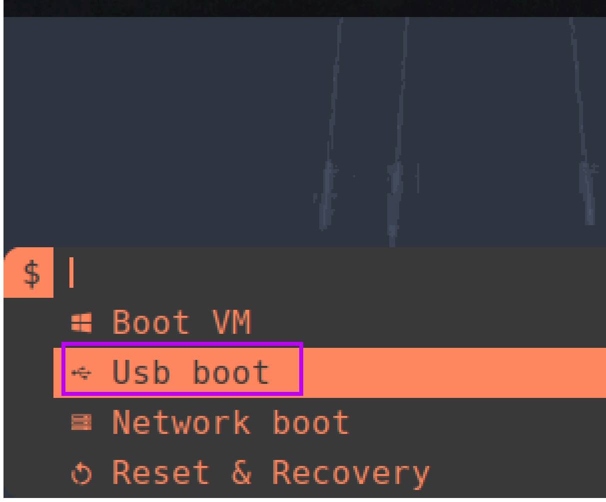


Other Features

Live USB Boot

A great feature of the VM toolset on p3ng0s is the support of running live USB directly from the machine to do that plugin in you flash drive and select the boot from USB option:





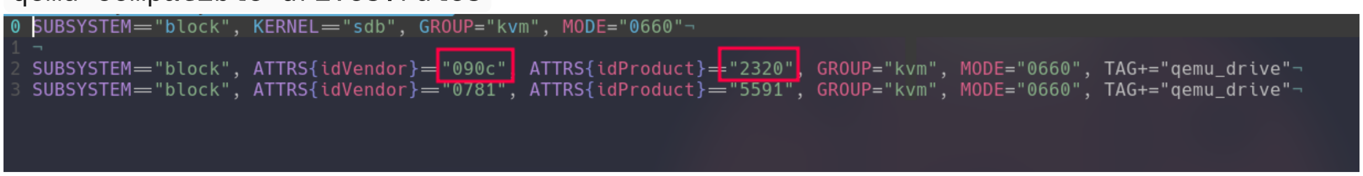
This feature is actually managed by udev and the `kvm` group. You need to first register a drive to be allowed to boot like this through: `/etc/udev/rules.d/97-qemu-compatible-drives.rules`

> **Important**
>
> Make sure your user is part of the `kvm` group

Adding a drive:

You first need the Vendor ID and Product ID of the drive you want to boot via `lsusb` command:

With those two numbers written down edit the following udev file: `/etc/udev/rules.d/97-qemu-compatible-drives.rules`



Simply add an entry like so

```
SUBSYSTEM=="block", ATTRS{idVendor}=="<VendorID>", ATTRS{idProduct}=="<ProductID>", GROUP="kvm", MODE="0660", TAG+="qemu_drive"
```

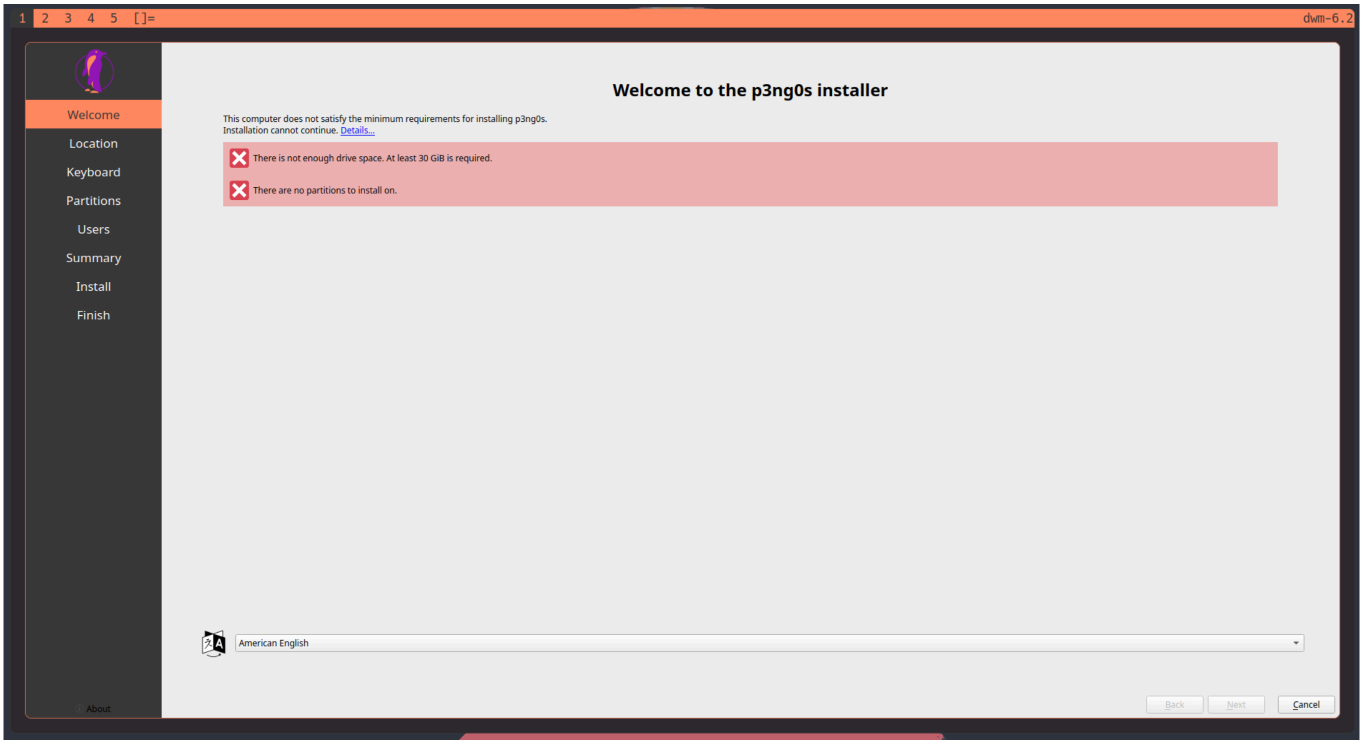
and reload the udev rules:

```
sudo udevadm control --reload && sudo udevadm trigger
```

You will then be prompted to select the drive you wish to boot form:

Booting

From there it will take a few seconds but you should be booting into your USB drive / Hard Drive that is plugged in to your machine:



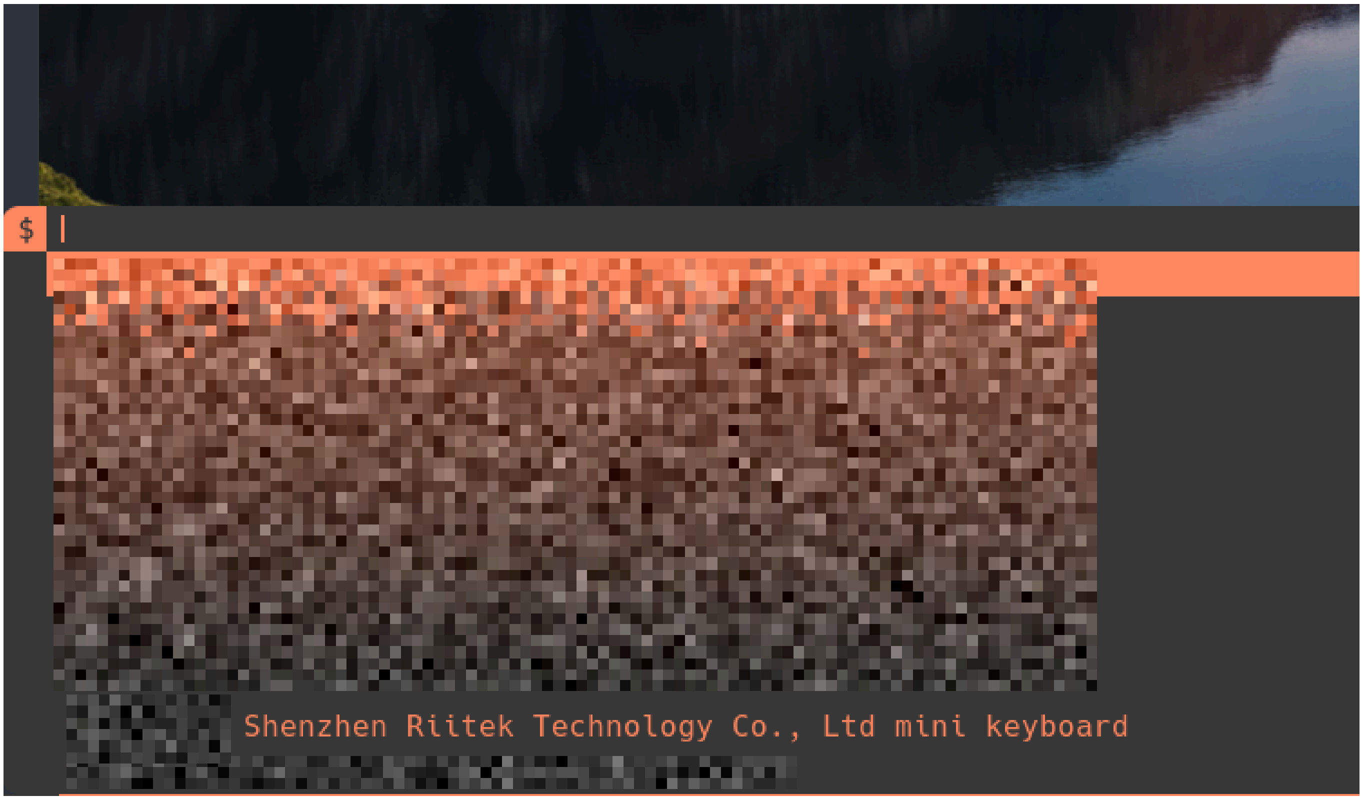
VM Management

USB Passthrough

Find your usb vendor id and product id through `lsusb` you can then add them to the following file `/etc/udev/rules.d/97-qemu-compatible-devices.rules` with the following syntax:

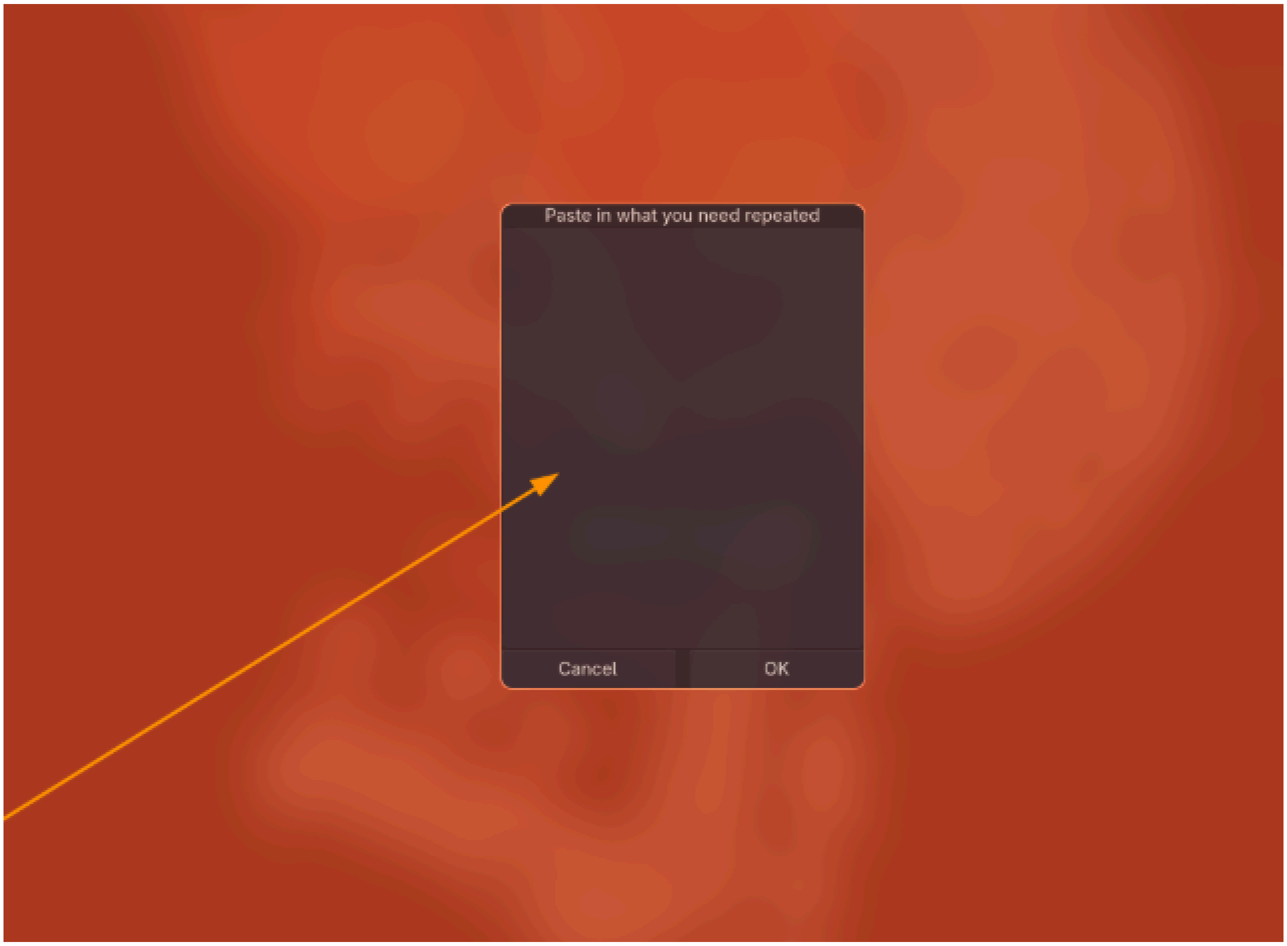
```
SUBSYSTEM=="usb", ATTRS{idVendor}=="<VendorID>", ATTRS{idProduct}=="<ProductID>", GROUP="kvm", MODE="0660", TAG+="qemu_usb"
SUBSYSTEM=="usb_device", ATTRS{idVendor}=="<VendorID>", ATTRS{idProduct}=="<ProductID>", GROUP="kvm", MODE="0660", TAG+="qemu_usb"
```

this will allow you to then while the VM is running passthrough a usb device with `Manage VM > Plugin USB`.



Paster

If you want to type in the VM a big string just select the `Paster` functionality and paste in whatever you want. Once you click Okay then the text will be auto typed into the VM make sure you have the right window selected:



Scripts

You can create scripts to automate things on the VM either type in text or cut off the internet of the VM.

Sample script:

```
string hello
set_link virtio-net-pci.0 off
wait 1
string world
```

Other

- Shutdown
- Reset
- Toggle Network ON/OFF

Useful commands to configure windows

Defender folder for compiling exploit code on windows

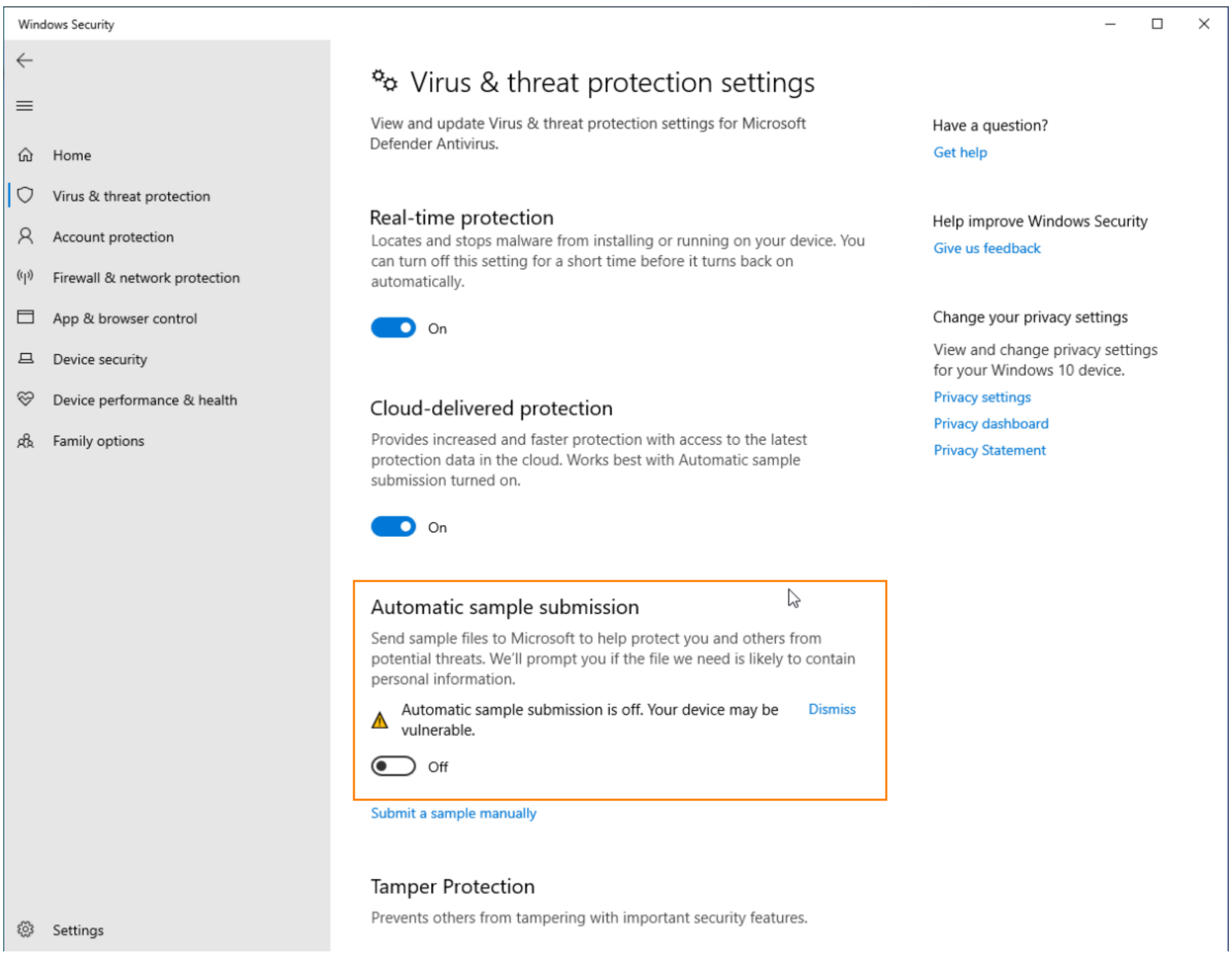
```
Add-MpPreference -ExclusionPath C:\Folder\Path
```

Find volumes

```
diskpart
list volume
exit
```

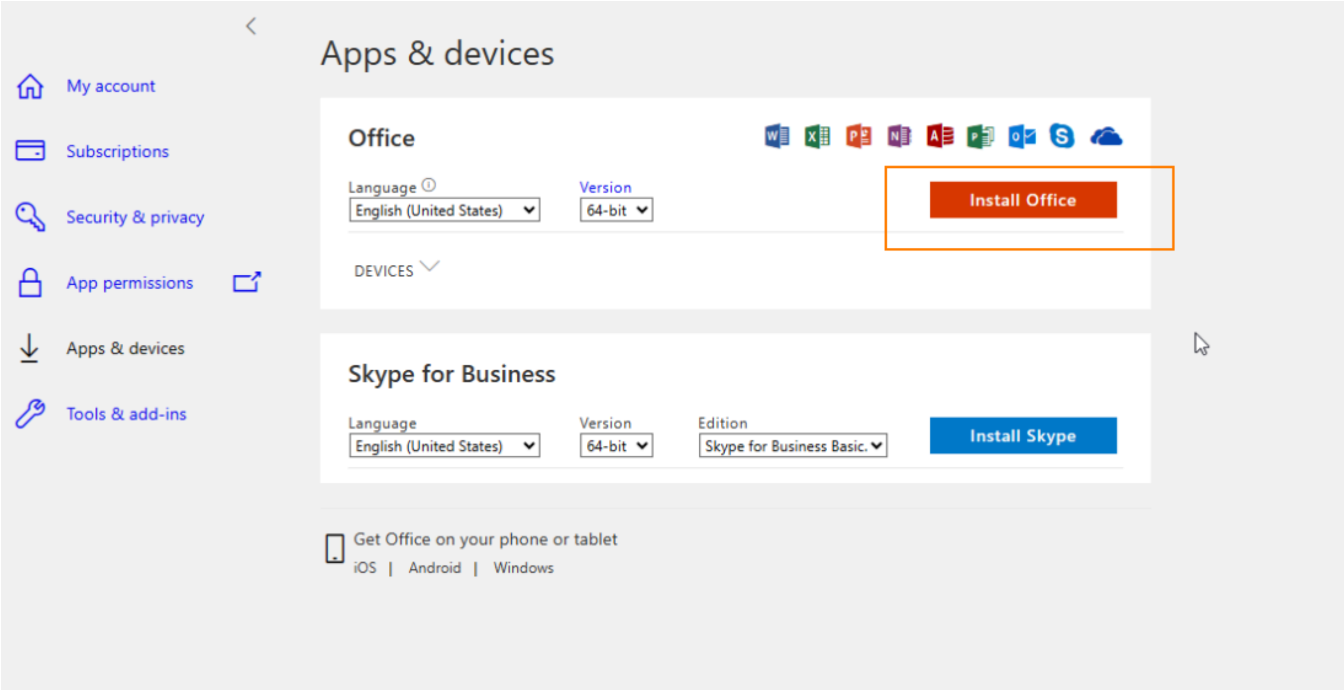
Defender settings for malware developement

You wan't to have all of defender active to check against the latest version of defender but it is important to disable the `Upload of malware` back to Microsoft so that if we trigger the AV our malware does not get uploaded back to Microsoft



Installing office

Go to `[aka.ms/office-install](http://aka.ms/office-install)` sign in then click on this:



Windows disable caps lock and set as escape

Map caps lock to escape in Windows | Vim Tips Wiki | Fandom

This tip shows some methods for mapping keys within the operating system to make life easier in Vim.
https://vim.fandom.com/wiki/Map_caps_lock_to_escape_in_Windows#Registry

```
Windows Registry Editor Version 5.00

[HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\Keyboard Layout]
"Scancode
Map"=hex:00,00,00,00,00,00,00,00,03,00,00,00,3a,00,46,00,01,00,3a,00,00,00,0
0,00
```

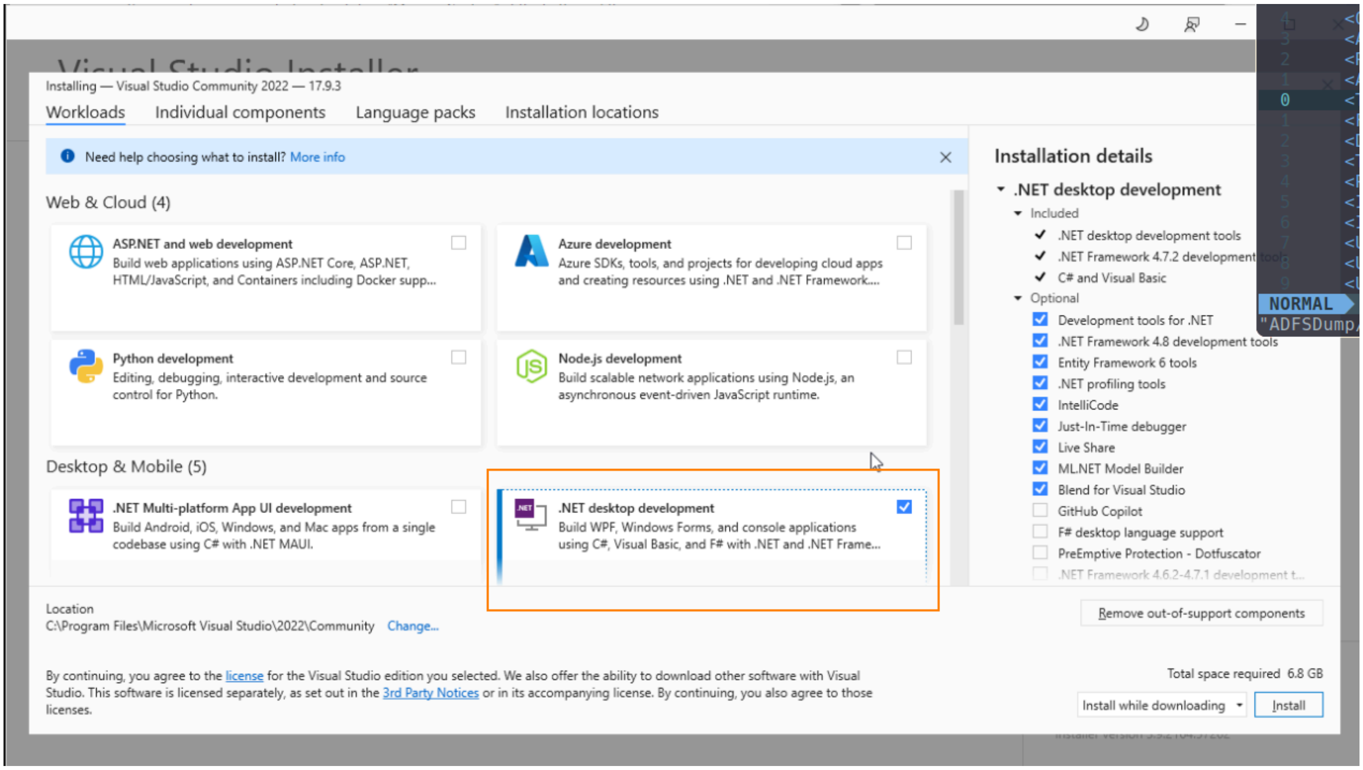
The MSBUILD problem

Download the comunity one:

Download Visual Studio Tools - Install Free for Windows, Mac, Linux

Download Visual Studio IDE or VS Code for free.
https://visualstudio.microsoft.com/downloads/

Once installing select this:



After going through the install you can find the MSBuild binary here:

```
&'C:\Program Files\Microsoft Visual
Studio\2022\Community\MSBuild\Current\Bin\MSBuild.exe'
```

Dissable git ssl check in corporate signed environements

```
GIT_SSL_NO_VERIFY=true git clone https://example.com/path/to/git
```

Useful windows tools list

Here is a collection of windows tools that can be interesting to install on your windows machine. This is mainly for me since I always forget the windows tools but maybe someone else out there can find this toolbox usefull as well:

Pentesting

- https://web.archive.org/web/20190101122212/http:/www.oxid.it/cain.html
- https://www.immunityinc.com/products/debugger/

Office sys admin suite

Sysinternals Suite - Sysinternals

The Windows Sysinternals troubleshooting Utilities have been rolled up into a single suite of tools.
https://learn.microsoft.com/en-us/sysinternals/downloads/sysinternals-suite

Chocolatey - The package manager for Windows

Chocolatey is software management automation for Windows that wraps installers, executables, zips, and scripts into compiled packages.
https://chocolatey.org/

Info

http://aka.ms/office-install

Malware Dev

x64dbg

An open-source x64/x32 debugger for windows.
https://x64dbg.com/

https://github.com/hasherezade/pe-bear

Downloads - Process Hacker

Process Hacker, A free, powerful, multi-purpose tool that helps you monitor system resources, debug software and detect malware.
https://processhacker.sourceforge.io/downloads.php